

*gain***observed t** **true $t+10$** **Direct error****AR error****ShiftWM error**#1
31%#2
31%#3
30%

per-patch error ($k = 10$), one scale for all panels; badge = frame mean
 $\text{gain} = 1 - \text{err}_{\text{ShiftWM}} / \min(\text{err}_{\text{Direct}}, \text{err}_{\text{AR}})$

 ≤ 0.04

0.60

 ≥ 1.16 **decoded truth****decoded Direct****decoded AR****decoded ShiftWM**

#1



#2



#3



*decoded $k = 10$
forecasts
(RGB decoder,
illustration only)*